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A Game-Changing Strategy for Modeling Fracture Reservoirs

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The simulation of fluid flow in fractured media can be approached through either single-continuum or dual-continuum models, with the latter often relying on the conceptual framework introduced by [Warren and Root \(1963\)](#). In both cases, a continuous representation of fracture properties—such as permeability, porosity, and connectivity—is required to enable numerical modeling at the reservoir scale ([Kazemi, H., et al., 1976](#)). However, obtaining such a continuous description directly from field data is particularly challenging, given the inherent heterogeneity and complexity of natural fracture systems.

This difficulty has led to the development of Discrete Fracture Network (DFN) models, which offer a more geologically grounded approach by explicitly representing individual fractures and their spatial organization. DFNs serve as a bridge between geological observations and flow simulation requirements: they allow for the generation of synthetic yet realistic fracture networks that can be upscaled to derive equivalent continuous properties for use in continuum models. Alternatively, DFNs can be used directly in flow simulations, particularly when the resolution of the model permits, or in geomechanical analyses where the discrete nature of fractures plays a critical role.

The contrast between continuous and discrete representations is thus central to the modeling of fracture reservoirs. While continuum models offer computational efficiency and are well-suited for large-scale simulations, they often rely on assumptions that may oversimplify the underlying fracture architecture. DFN models, on the other hand, provide a more detailed and mechanistic view of fracture systems, albeit at the cost of increased computational demand and data requirements. The integration of both approaches remains a key challenge and opportunity in the field of fractured reservoir modeling.

Introduction

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However, despite the conceptual and practical advantages offered by DFN models, they remain far from perfect. Significant limitations persist, particularly in their ability to fully capture the geological and mechanical complexity of natural fracture systems. This paper aims to address these shortcomings by proposing substantial improvements to current DFN modeling practices.

The first part of the paper provides a critical review of the DFN modeling framework, highlighting its foundational principles and the major limitations that arise from the oversimplification - or complete omission - of fracture typologies and hierarchical organization. It also examines recent methodological advances that attempt to overcome these issues, while emphasizing the persistent gaps that remain even in the most advanced commercial software.

The second part presents a set of very new solutions packaged in a proprietary modeling tool, designed to integrate geomechanical and geological rules directly into the DFN workflow. This tool emulates successive phases of fracture development and manages various scales and types of faults and joints, generated according to deformation style and consistent with stratigraphy. It also offers a more robust handling of fracture relationships and upscaling procedures. Despite its remarkably simplified parameterization compared to most commercial software, this in-house solution delivers a very consistent pathway from discrete fracture modeling to flow simulation, demonstrating that modeling efficiency and scientific accuracy can indeed coexist.

DFN models: the question of fracture typology

A cornerstone in the building of fracture networks

The foundation of DFN modeling lies in the explicit description of fractures as discrete and geometrically defined objects. Since it has been proposed, this strategy has been successful, and over the last decades, DFN models have become a cornerstone in the building of fracture networks for characterizing and simulating fluid flows in fracture reservoirs. [Dershowitz and Einstein \(1988\)](#) pioneered the use of statistical distributions to describe fracture properties such as orientation, size, and density. Today, each fracture is characterized by its orientation, shape, dimensions, aperture, and spatial location, allowing for a more realistic representation of the reservoir's heterogeneity. The statistical distributions of fracture properties used by DFN models are classically derived from field data, and possibly calibrated on log data, borehole images, core samples, seismic attributes, or outcrop analogs. This explicit representation is critical for enabling DFN to determine the connectivity and permeability of the network ([Priest, 1993](#)), to capture the complex interplay between fractures and the matrix, and therefore to provide a more accurate depiction of fluid flow pathways ([Adler & Thovert, 1999](#)).

The Importance of Fracture Typology in Geological Systems

Despite their widespread adoption and technological advancements, DFN models are not without substantial limitations, particularly in their ability to capture the full geological complexity of fracture systems. One of the most critical, yet often overlooked, limitations of DFN models is their inability to explicitly distinguish between different fracture types, such as joints, faults, fault arrays and fracture clusters. However, the distinction between fracture types is crucial for accurate fluid flow modeling. Joints, with their pervasive and interconnected nature, often facilitate fluid movement across large volumes of rock. Faults, depending on their internal architecture, can either enhance or impede fluid flow, while fracture clusters and shadow zones introduce additional complexity by creating preferential flow paths or barriers.

The failure to explicitly represent fracture types in DFN models can lead to oversimplified representations of geological structures, where the unique mechanical, hydraulic, and geometric properties of distinct fracture types are not adequately captured, with a potential misunderstanding of the true anisotropy and heterogeneity of the fracture network. Consequently, DFN models may fail to accurately reflect the true behavior of fracture reservoirs, leading to potential inaccuracies in fluid flow predictions and reservoir management strategies.

What is fracture typology?

Fractures in geological formations are not homogeneous entities; rather, they exhibit a wide range of types, each with distinct characteristics and implications for fluid flow. Broadly, fractures can be categorized into several types (D.C.P. Peacock et al., 2016) including joints, faults, and fracture clusters, each of which plays a unique role in controlling the mechanical and hydraulic behavior of the subsurface reservoirs.

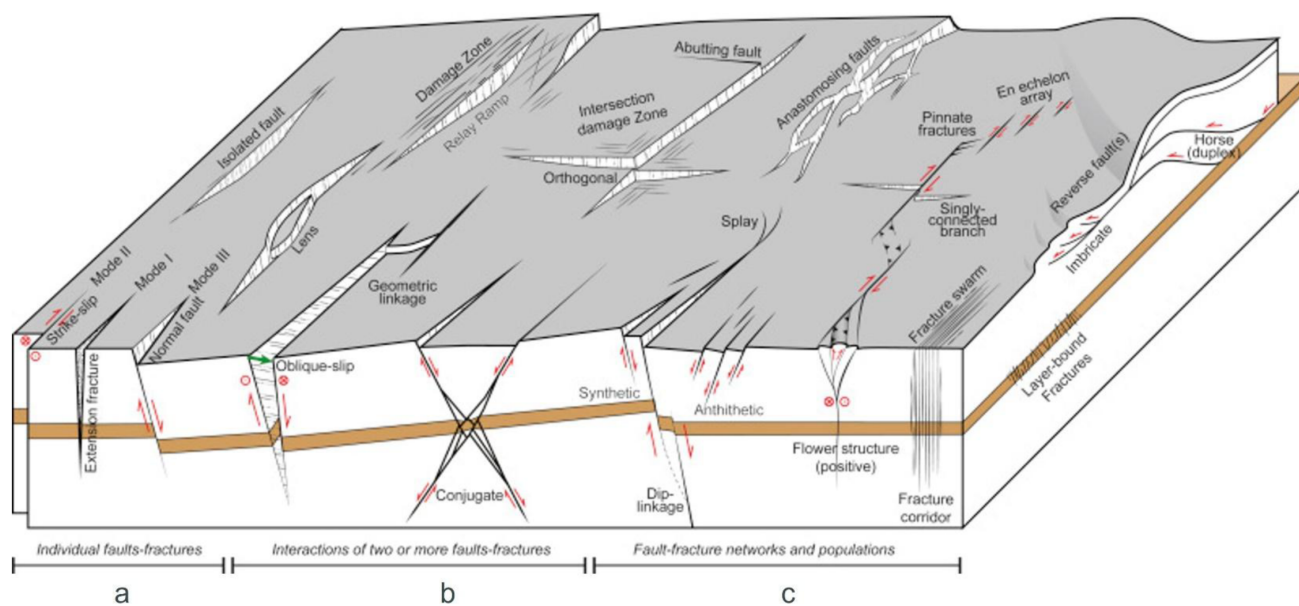


Figure 1—Schematic illustration of the range of fault and other fracture networks, after D.C.P. Peacock et al. (2016). (a) Individual fractures; (b) Interaction between pairs of fractures; (c) Fracture networks.

Joints. Joints are planar fractures that form due to tensile or shear stresses and typically exhibit minimal displacement. They are often pervasive in sedimentary rocks and can significantly enhance permeability by providing interconnected pathways for fluid flow (Pollard & Aydin, 1988). Joints are commonly organized into systematic sets, with consistent orientations that reflect the regional stress field at the time of their formation. The presence of multiple joint sets can create a highly anisotropic permeability structure, where fluid flow is preferentially oriented along certain directions (Nelson, 2001).

Faults. Faults, in contrast, are fractures along which significant displacement has occurred. They can act as either conduits or barriers to fluid flow, depending on their internal structure and the nature of the fault gouge (fine-grained, clay-rich zone formed by intense mechanical grinding and chemical alteration along a fault). It typically exhibits low permeability and can act as a barrier or seal to fluid flow within geological formations. Fault zones often comprise a complex architecture, including a core of highly deformed rock surrounded by a damage zone characterized by an increased density of smaller fractures (Caine et al., 1996). The hydraulic properties of faults are highly variable and can evolve over time due to processes such as mineral precipitation, fault reactivation, and stress-induced aperture changes. As such, faults play a critical role in compartmentalizing reservoirs and influencing fluid migration pathways (Faulkner et al., 2010).

Fault Arrays and Fracture clusters. Faults and fracture networks refer to clusters or zones of fractures that exhibit a consistent orientation or spatial distribution, often associated with specific geological features such as folds, shear zones, or lithological boundaries. These arrays can create highly permeable corridors that dominate fluid flow within a reservoir.

Current representation of fracture types in DFN models

In traditional DFN modeling, fractures are typically represented as planar, two-dimensional surfaces with defined geometric properties such as orientation, size, and aperture. Basically, the main parameters which drive 3D organization of fractures in DFN models are related to statistics, as distributions of parameters, values of fracture densities, or coefficients evaluating the degree of fracture clustering. In the very classic stochastic modeling of fractures, geomechanics is lacking, whether through mechanical calculations or the rules that structural geology could provide. Consequently, DFN models often treat all fractures as equivalent entities, characterized by similar mechanical and hydraulic properties. This homogenization of fracture properties can lead to significant inaccuracies, particularly in reservoirs where different fracture types coexist and interact. For example, a DFN model that does not distinguish between joints and faults may overestimate or underestimate permeability, depending on the relative abundance and connectivity of each fracture type (Dershowitz et al., 2000).

Challenges in Incorporating Fracture Typology

The primary challenge in incorporating fracture typology into DFN models lies in the complexity of geological data and the limitations of current modeling techniques. Field data often provide insufficient resolution to distinguish between different fracture types, particularly in subsurface environments where direct observations are limited. Additionally, the mechanical and hydraulic properties of fractures are influenced by a multitude of factors, including stress history, lithology, and diagenetic processes, which are difficult to capture in a simplified modeling framework (Laubach et al., 2019).

Another challenge is the computational demand associated with explicitly representing multiple fracture types. Each type may require distinct geometric and hydraulic parameters, increasing the complexity of the model and the computational resources required for simulation. This complexity is further compounded by the need to account for interactions between different fracture types, such as the termination of joints against faults or the development of shadow zones in the vicinity of major fractures (Odling et al., 1999).

New advanced Solutions

Recently, the development of more sophisticated numerical methods and computational algorithms have tried to enable the explicit representation of multiple fracture types, improving the accuracy and predictive capabilities of DFN models. These softwares incorporate new fracture generation procedures and advanced meshing, allowing for the explicit representation of fracture sets and their properties. This enables users to build models that better reflect the heterogeneity and connectivity of fracture networks, including the distinction between joints and faults, thanks to a couple of new approaches:

The hybrid approach. It allows for a more nuanced representation of the fracture network, capturing both the heterogeneity of individual fractures and the overall connectivity of the system (Karimi-Fard et al., 2004). With such a technique, large-scale features can be explicitly represented using DFN, while smaller-scale fractures are accounted for using continuum approaches. This hybrid methodology helps balance computational efficiency with geological realism, allowing for more accurate simulations of fluid flow and mechanical behavior in complex reservoirs.

The coupled hydro-mechanical modeling. Some DFN tools increasingly support coupled hydro-mechanical simulations, allowing for the dynamic representation of fracture behavior (e.g., fault reactivation, stress-dependent permeability). This capability is critical for accurately modeling the interaction between different fracture types and their impact on fluid flow and reservoir mechanics.

The multi-scale fracture modeling. The solution proposed by one of the DFN tools is to represent fracture systems from regional faults down to sub-seismic and micro-fractures, maintaining a certain geological realism. Diffuse fractures (stratabound and non-stratabound) are distinguished from faults (deterministic and stochastic) in the modeling workflow. By defining fracture sets per stratigraphic unit, fracture models are stratigraphically constrained and geologically consistent.

Remaining critical gaps in the geological representation of fracture networks

Although the most advanced software distinguishes between faults and joints, the fracture networks generated by most DFN modeling tools still offer a rather simplistic representation of the true geological complexity of these systems. Several critical aspects remain either insufficiently addressed or entirely absent from most commercial tools, highlighting the need for further development.

Even in the most advanced software, while a distinction is made between faults and joints, the explicit representation of the various scales and specific organizational patterns of fracture networks remains largely unattainable. Many key parameters that govern the spatial arrangement and connectivity of fractures are often overlooked and replaced by statistical proxies such as density metrics or value distributions, which fail to capture the underlying geological controls.

Furthermore, the concept of tectonostratigraphy is only superficially addressed in a few tools. In particular, the role of mechanical stratigraphy—i.e., the presence of mechanically distinct units at different scales—is generally neglected in modeling workflows, despite its fundamental influence on fracture development and propagation.

The parametrization of fracture density and geometric properties across different fracture types also presents a significant challenge for the average user. The complexity involved in defining these characteristics often leads to oversimplified models that do not reflect the heterogeneity observed in the field.

As well, one of the persistent limitations in current DFN modeling approaches lies in the inadequate representation of the complex interactions between different types of fractures. In natural systems, fractures rarely exist in isolation; they intersect, abut, deviate, and influence each other in ways that are both geometrically and mechanically significant. These relationships—such as cross-cutting patterns, termination behaviors, and reactivation histories—play a crucial role in controlling fluid flow and mechanical responses within the reservoir. When DFN tools treat fractures as independent entities, interaction between fractures is lacking or poorly represented. This simplification not only affects the geological realism of the models but also undermines the accuracy of fracture connectivity and of subsequent flow simulations.

Finally, DFN models are frequently developed with the sole purpose of upscaling, as flow simulations, whether in single or dual-porosity media, require the computation of equivalent network properties. This upscaling phase is typically carried out using either analytical or numerical methods. However, the outcome

is most often a homogenized equivalent matrix medium, with few alternatives offered to represent the discrete nature of the original fracture system.

Improving Geological Realism in DFN Models

A rule-based DFN model

A proprietary forward fracture modeling tool has been created using a rule-based approach. The application is intended to replicate geological and geomechanical processes without directly solving physical equations. Rather than relying on direct numerical simulation of stress and strain fields, the software operates through a set of predefined rules that govern the relationships between key descriptive parameters of fracture systems. These rules are expressed mathematically, enabling the generation of geologically plausible fracture networks while maintaining a high degree of user control and interpretability.

A central concept is the use of geological time as a structuring axis for the modeling workflow. The deformation history of the reservoir is divided into successive phases, each corresponding to a distinct tectonic event or stress regime. For each phase, a specific fracture system is defined, characterized by its own geometry, orientation, and mechanical behavior. This temporal segmentation allows the model to reflect the cumulative and hierarchical nature of fracture development in sedimentary basins.

To simplify the modeling process for users, the software provides a library of predefined fracture types associated with each deformation phase. These include various families such as fault, fault arrays, bed-confined fractures, and fault-related damage zones. By associating each system with a set of typical fracture geometries and propagation rules, the tool reduces the complexity of parameterization while enhancing geological realism.

Notably, the application maintains a stochastic element, thereby preserving variability and uncertainty in both fracture characteristics and their spatial distribution. Although the model is governed by a set of rules, it does not aim to replicate the complete sequence of physical processes responsible for fracture formation. Instead, it mimics these processes through probabilistic and rule-based logic, offering a balance between geological plausibility and computational efficiency.

This hybrid approach—combining deterministic rules with stochastic variability—enables this software to simulate not only the spatial distribution of fractures but also their evolution over time, including interactions between successive systems. The resulting models can be used for flow simulations—either after upscaling to continuum properties or directly on the discrete objects—as well as for geomechanical analyses. By embedding geological and mechanical logic into the modeling workflow, the application offers a flexible and robust framework for representing complex fracture networks in subsurface environments.

Fracture Systems and fracture typology

The proprietary rule-based algorithms are designed to reflect the structural logic and mechanical behavior associated with fracture systems, as defined in the typology. Each fracture system is associated with a specific deformation phase, which is defined along a geological time axis. During each phase, the user selects a deformation style (e.g., extensional, compressional, strike-slip), which activates a corresponding set of rules and fracture types. These rules are not purely geometric; they incorporate stratigraphic and mechanical constraints, ensuring that fractures are placed in geologically plausible locations and orientations.

The classification of fracture systems is structured around tectonic regimes and the characteristic deformation styles they produce. This typology serves as a foundation for rule-based modeling, allowing the software to associate specific fracture geometries with the mechanical context in which they typically form. Four principal fracture systems are defined, each encompassing a distinct set of fracture types.

Table 1—Deformation systems incorporated within the software, along with corresponding objects or object combinations that may be produced during a tectonic phase. For example, for a Mode I deformation, which results in the initiation and propagation of systematic joint systems, three types of joints are generated by the software: longitudinal joints, cross joints, and oblique joints.

Fracture System	Fracture Type
Systematic Joints System	Longitudinal Joints
	Cross Joints
	Oblique Joints
Normal Dip-Slip Faults & Extensional System	Anastomosing Faults
	Relay Ramp Faults
	Faults Lineaments
	Isolated Faults
	Fracture Corridors
Reverse Dip-Slip Faults & Compressional System	Faults Lineaments
	Isolated Faults
Strike-Slip Faults System	Anastomosing Faults
	En Echelon Faults
	Faults Lineaments
	Isolated Faults
	Fracture Corridors

The first system, referred to as the systematic joints system, develops in Mode I conditions and includes fractures that develop under relatively uniform stress conditions, often in undeformed or gently deformed sedimentary rocks. Within this system, longitudinal joints align with the principal stress direction, cross joints form perpendicular to them, and oblique joints appear at intermediate angles, reflecting local variations in stress orientation. These joints are typically stratabound and represent early-stage or low-intensity deformation features.

The second system corresponds to normal dip-slip faults and extensional tectonics. It includes anastomosing faults, which form braided networks with complex connectivity, as well as relay ramp faults that characterize segmented fault systems with transfer zones. Fault lineaments and isolated faults are also part of this system, representing linear and discrete fault expressions respectively. Fracture corridors, which are zones of concentrated fracturing, are particularly significant in this context due to their role in enhancing permeability and controlling fluid flow.

The third system is associated with reverse dip-slip faults and compressional tectonics. It comprises fault lineaments and isolated faults, typically steeply dipping and mechanically constrained by stratigraphy. These features contribute to structural compartmentalization and may act either as barriers or conduits depending on their orientation and sealing properties.

Finally, the strike-slip fault system includes a range of shear-related features. Anastomosing faults and en echelon faults reflect distributed shear and progressive deformation, while fault lineaments and isolated faults represent major strike-slip structures and localized responses to stress. Fracture corridors also appear in this system, often aligned with principal shear directions and contributing to anisotropic flow behavior.

This structured typology enables the simulation of fracture networks that are geologically consistent and mechanically plausible. By linking fracture types to tectonic systems, the model provides a coherent framework for representing the evolution of fractures across successive deformation phases, facilitating their integration into both flow simulation and geomechanical analysis workflows.

A simplified parameterization

A key advantage of the proprietary software is its efficient and simplified approach to parameterization, distinguishing it from the typically intricate and time-consuming workflows present in numerous commercial fracture modeling solutions. This solution is designed to minimise user input while substantially improving geological authenticity.

At the heart of this simplification is a rule-based system that automatically derives a wide range of fracture attributes from a limited set of user-defined inputs. For example, when the user specifies the strike azimuth of a given fracture system, the software automatically calculates the strike and dip azimuths for all fracture types associated with that system. This eliminates the need for manual input of orientation parameters for each individual fracture family, significantly reducing the cognitive and operational load on the user.

The mechanical rules embedded in the software further enhance this efficiency. These rules encode geological and structural logic, allowing the software to infer relationships between parameters such as fracture spacing, density, and object dimensions. As a result, the number of parameters that must be explicitly defined by the user is reduced by more than a factor of ten compared to traditional workflows. Despite this automation, users retain full control over the modeling process and can override or edit any parameter as needed, ensuring flexibility and transparency.

In addition to orientation and geometry, the programme also automates the calculation of density and spacing parameters, as well as the scaling relationships between fracture dimensions. These calculations are dynamically adjusted based on the selected deformation style, the types of fracture objects to be generated, and the desired resolution of the model. This adaptive logic ensures that the resulting fracture networks are consistent with both geological expectations and modeling objectives.

By incorporating geological insight into the parameterization process, this modeling programme offers both user accessibility and a strong foundation in scientific rigor. It enables rapid construction of complex fracture networks without sacrificing interpretability, making it an ideal tool for both expert users and those less familiar with the intricacies of fracture mechanics.

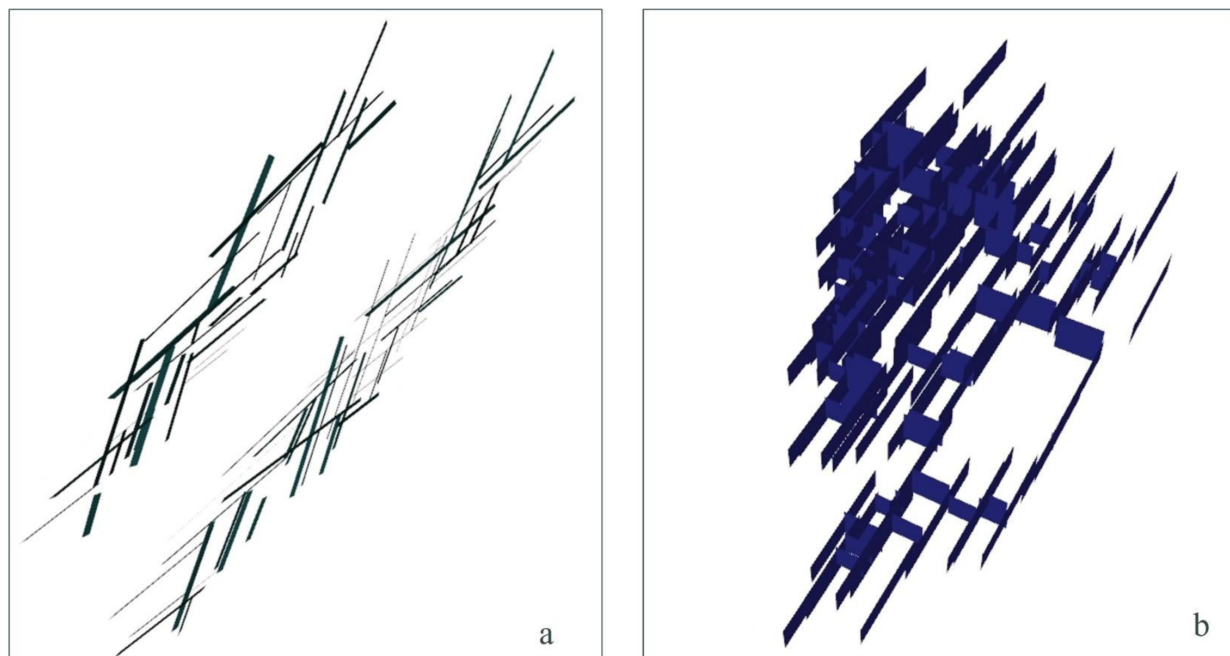


Figure 2—Two types of fault arrays generated "naturally" in a stochastic manner, without manipulating densities or other statistical parameters. The type of array must be only chosen by the user in the library of available objects. (a) Anastomosed faults; (b) Relay ramp faults.

Mechanical and fracture stratigraphy

A key innovation lies in the explicit integration of mechanical and tectonic stratigraphy into the fracture modeling workflow. Rather than treating the stratigraphic column as a homogeneous or purely geometric framework, the software distinguishes between elementary stratigraphic units (individual beds) and mechanical units, which group beds with similar mechanical behavior. This dual stratigraphic structure allows the model to simulate fracture development in a way that reflects both sedimentary architecture and tectonic history.

This framework is particularly powerful for generating stratabound fractures, which are confined to specific stratigraphic or mechanical units. The software supports the generation of such fractures across a wide range of deformation styles and fracture types. For fault-related features, the software can simulate stratabound fault segments that are mechanically constrained by lithological contrasts or bedding planes. More notably, in the case of joints, the software offers extensive capabilities to reproduce the full typology of fractures defined by L. Bazalgette (2004), including Bed-Controlled Fractures (BCF), Multiple Beds Fractures (MBF), Highly Persistent Fractures (HPF), and Joint Swarms (JS).

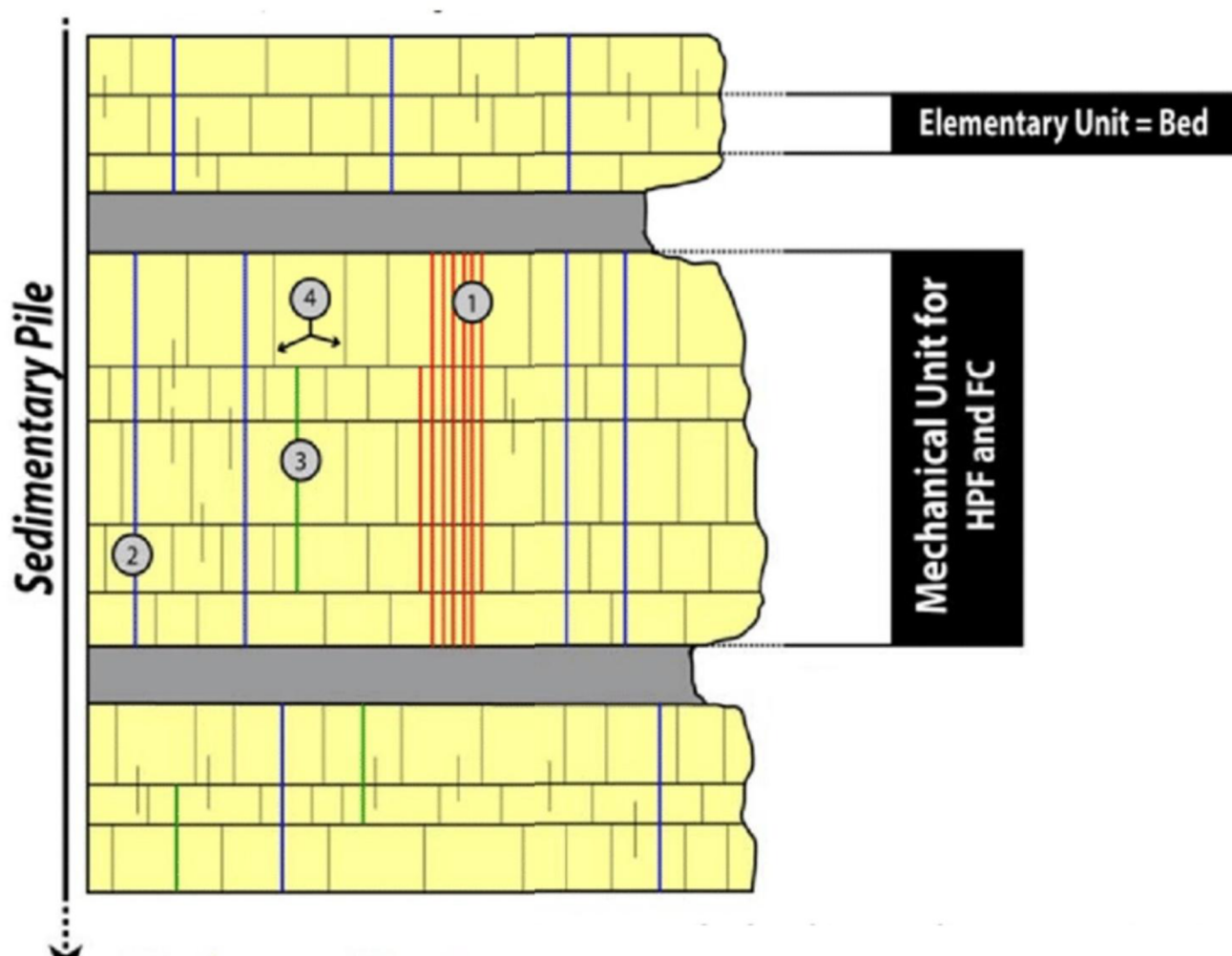


Figure 3—Relationship between scales of jointing and mechanical units of varying thicknesses (modified after L. Bazalgette, 2004). The in-house software typically reproduces such an organization between the different scales of systematic joints, with a parameterization of spacing automatically computed from bed rheology, bed thickness and intensity of deformation. (1) FC/JW: Fracture Corridors/Joint Swarms are clusters of joints limited to the mechanical unit; (2) HPF: Highly Persistent Fractures cross the whole mechanical unit; (3) MBF: Multiple Beds Limited Fractures cross several beds but they still are limited by bed interfaces; (4) BCF: Bed Controlled Fractures are the smallest systematic joints;

Each of these fracture types is associated with specific rules that govern their geometry, orientation, and spatial extent. BCFs are confined to individual beds and are influenced by sedimentary texture and local mechanical properties. MBFs span several beds within a single mechanical unit, reflecting intermediate-scale deformation. HPFs are long, continuous fractures that propagate across a single mechanical unit, often associated with regional tectonic phases. Joint swarms represent dense, localized clusters of fractures, typically formed under uniform stress conditions and stratigraphically constrained.

The rule-based architecture of the proprietary software ensures that these fracture types are generated in geologically plausible locations, with orientations and dimensions that reflect both the stratigraphy and the deformation style selected by the user. The software automatically adjusts parameters such as fracture spacing, density, and extent based on the mechanical context and resolution of the model, while still allowing the user to override or refine these settings.

The software incorporates mechanical and tectonic stratigraphy into its core logic, allowing for the generation of fracture networks that maintain structural coherence and stratigraphic consistency. This capability is essential in layered reservoirs, where the interaction between fractures and stratigraphy plays a critical role in controlling connectivity and permeability.

Fracture interactions

A notable advanced capability of the programme is its capacity to accurately replicate the intricate relationships among fractures, as identified through field observations or deduced from seismic interpretation. Accurately replicating geometric, chronological, and mechanical relationships is crucial in DFN models. The application addresses this complexity by utilising rule-based algorithms, temporal sequencing, and energy-based interaction frameworks.

The software allows users to define and control abutting relationships between different fracture types and scales, particularly among joints. These interactions are fully parameterized, enabling the model to simulate realistic termination behaviors, cross-cutting patterns, and hierarchical organization. Similarly, the relationships between joints and faults are explicitly managed, ensuring that joint systems respond appropriately to fault propagation and orientation.

In addition to joint-fault interactions, the modeling also includes relationships among different fault types, fault arrays, and the connections between faults and larger fracture systems. These relationships can be governed either by the chronology of deformation phases, which reflects the geological history of the reservoir, or—at the user's discretion—by a surface-based energy comparison, which evaluates the relative mechanical influence of each object. This dual approach allows for both geologically driven and physically inspired modeling strategies.

A key aspect of this functionality is the guidance provided to the user. Rather than requiring the user to manually define complex interaction rules - often based on data that may be unavailable for a given field - the solution embeds academic and global knowledge into its rule sets. This approach guarantees that users benefit from evidence-based defaults and recommendations, thereby minimising the likelihood of oversimplification or inaccurate representation. In some cases, the software may even constrain user choices to maintain geological plausibility, while still allowing for expert overrides when justified.

The development team is also actively working on a prototype for a feature that will allow for local adjustment of the stress field. This functionality enables the adjustment of deformation intensity in specific areas and the control of how subsequent fracture objects propagate. Early tests have shown promising results, suggesting that this capability could significantly enhance the realism and adaptability of fracture models, especially in structurally complex or heterogeneous reservoirs.

Through its comprehensive management of fracture relationships, this rule-based DFN model offers a modeling environment that is both geologically informed and operationally efficient. This approach facilitates the integration of theoretical knowledge with practical implementation, allowing users to develop fracture networks that maintain structural integrity while adapting effectively to geological dynamics.

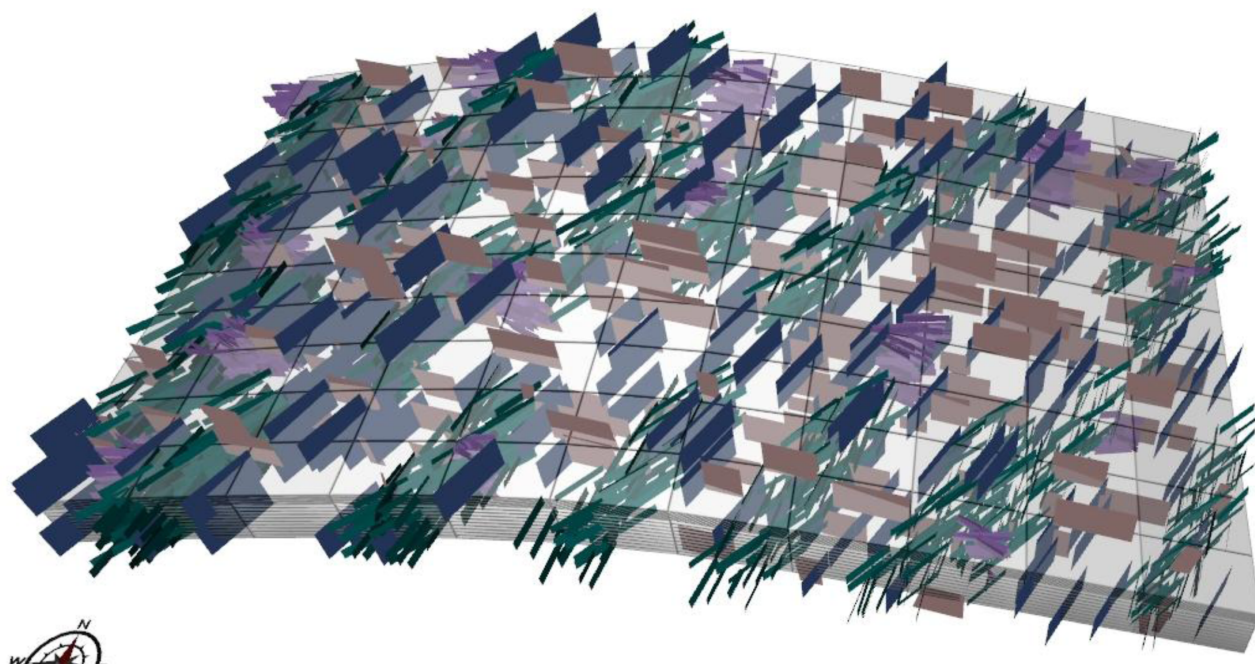


Figure 4—A very simple model populated only with large scale faults. Two successive fracture systems are composed of anastomosed and isolated faults, the second fracture system abutting on the first one.

Upscaling the DFNs: Bridging the Discrete and the Continuous

Connectivity is a key parameter in DFN modeling, as it directly influences the reservoir's permeability and fluid flow behavior. Fracture connectivity is assessed using topological analysis, which evaluates the degree to which fractures are interconnected. Highly connected networks facilitate fluid flow, while poorly connected networks can lead to compartmentalization and reduced recovery efficiency (Berkowitz et al., 2000). Permeability in DFN models is often calculated through upscaling techniques which aim to represent the effects of small-scale fractures on larger-scale flow behavior, enabling the integration of DFN models with conventional reservoir simulators. This integration is essential for practical applications, where the detailed representation of fractures must be balanced with computational feasibility (Durlafsky, 1991).

In the context of fractured reservoir modeling, upscaling remains an essential yet conceptually challenging step, particularly when transitioning from detailed discrete fracture representations to continuous equivalent properties suitable for flow simulation. This transition often involves a form of "quantum leap", a radical simplification in which the rich geometric and topological complexity of DFNs is reduced to a homogenized medium. While necessary for computational tractability, this process risks losing critical information about connectivity, anisotropy, and localized flow behavior.

Our software addresses this challenge by offering a suite of upscaling methodologies, both analytical and numerical, each tailored to different modeling needs and fracture network characteristics. The analytical approach is based on the well-known Oda method (1985), which computes an equivalent permeability tensor from the statistical orientation and aperture of fractures. This method is computationally efficient and well-suited for highly connected networks. However, its limitations become apparent in sparse or poorly connected systems. To overcome these limitations, the software integrates a modified version of the Oda method, inspired by the work of Rajeh et al. (2019), who extended the analytical framework to incorporate matrix contributions into the equivalent medium. This enhancement allows for a more realistic representation of dual-porosity systems, where both fractures and matrix play a role in fluid transport. The modified method is designed to address accuracy in heterogeneous reservoirs and offers an alternative perspective on permeability distribution. It is automatically activated when the matrix is present.

In parallel, the modeling tool includes a numerical upscaling module, built around a in-house high-performance pressure solver. This method simulates flow directly through the discrete fracture network and computes equivalent properties based on actual pressure responses. While more computationally intensive, numerical upscaling offers superior accuracy in complex or poorly connected networks and is particularly valuable when analytical assumptions break down or when dissolution conduits are involved in the upscaling.

Recognizing the strengths and limitations of each approach, our software does not impose a single upscaling strategy. Instead, it provides users with the flexibility to choose between equivalent matrix representations, equivalent fracture representations, or direct modeling within conductive zones. For example, fracture corridors, fault damage zones, and joint swarms can be upscaled as discrete conductive features rather than being absorbed into a homogenized matrix. This hybrid modeling capability allows users to preserve structural detail where it matters most, while simplifying other regions for computational efficiency.

To support this flexibility, the upscaling procedure also includes original algorithms for computing matrix-fracture exchange, a critical parameter in hydrocarbon recovery. These calculations account for the geometry, spacing, and connectivity of fractures, as well as the local pressure gradients, enabling more accurate simulation of fluid transfer between domains.

By combining analytical and numerical methods, and by offering a modular and user-guided upscaling framework, this new in-house software provides a powerful and adaptable solution to one of the most difficult challenges in fractured reservoir modeling. It empowers users to maintain geological fidelity while meeting the practical demands of simulation workflows.

Conclusions

Such rule-based DFN modeling software represents a significant advancement in the modeling of fracture reservoirs, offering a paradigm shift in both methodology and usability. By combining rule-based systems with geological and mechanical knowledge, the software creates fracture models that preserve geological accuracy and improve business efficiency. Its simplified parameterization, far less demanding than that of most commercial tools, does not compromise accuracy; on the contrary, it delivers superior and more realistic results for flow simulations, demonstrating that modeling rigor and accessibility can coexist.

Throughout this article, the core innovations included in the software have been highlighted: its capacity to manage fracture typology according to deformation style, its integration of mechanical and tectonic stratigraphy, its ability to generate stratabound fractures across multiple deformation styles, its management of complex fracture relationships, and its flexible upscaling framework that bridges discrete and continuous representations. These features collectively empower users to build models that reflect the true complexity of subsurface systems, while maintaining control and interpretability.

Yet, not all available capabilities have been covered in this paper. Among the most promising developments is the solution to a long-standing challenge in reservoir simulation: the representation of fault zones with impermeable core zone and highly permeable damage halos. This dual behavior, difficult to capture in conventional simulators, is explicitly addressed in the software through hybrid modeling strategies that preserve structural detail where it matters most. In addition, the software includes original approaches for integrating seismic information and calibrating fracture models against dynamic data, further enhancing its predictive power.

These features open the door to a wide range of applications, particularly in complex carbonate reservoirs such as those found in the Middle East, where fracture architecture plays a decisive role in hydrocarbon recovery. This software is not merely an incremental improvement. It is a game-changing tool that redefines how fracture modeling can be approached. By incorporating geological expertise within a structured, adaptable workflow, this approach establishes an advanced benchmark for constructing high-

quality reservoir models. Developed as part of a specialized tool suite for geological forward modeling of reservoirs, this innovative methodology is anticipated to significantly influence the development and production strategies for fractured reservoir systems.

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